final_project

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1 Multi-Agent Financial Analysis System

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This notebook is the canonical, executable report for an agentic investment-research system. It accepts a stock ticker, plans a bounded research run, gathers market, fundamental, filing, and news evidence, routes news to domain specialists, evaluates and improves analysis, synthesizes a final report, and stores reusable lessons for later runs.

Repository: AAI-520 Final Project

> Research support only. The output is not personalized investment advice and does not guarantee future performance.

1.1 1. Project objectives and implemented capabilities

The project demonstrates four autonomous agent functions:

1. **Plan** research objectives, questions, and tool selection using prior-run memory as context.
2. **Use tools** to retrieve Yahoo Finance market/fundamental data, SEC EDGAR facts and filings, and multi-source company news.
3. **Reflect and evaluate** data quality, report grounding, and specialist-analysis completeness.
4. **Learn across runs** by persisting bounded, timestamped research lessons.

It also implements all three required workflow patterns:

- **Prompt chaining:** ingest → preprocess → classify → extract → summarize.
- **Routing:** earnings, market, and general-news articles go to matching specialists.
- **Evaluator–optimizer:** a four-part, 100-point rubric drives bounded revisions.

The refactored modules preserve the original notebook's planning, tool execution, validation, reflection, reporting, memory, SEC EDGAR retrieval, 10-year price graph, multi-source news analysis, specialist routing, and iterative optimization. The updated Google News RSS fallback and retry-aware HTTP behavior are included as well.

1.2 2. Implemented system architecture

The research workflow keeps provider adapters, provider-independent tools, agents, workflows, memory, and presentation concerns separate. This makes each stage inspectable and independently testable.

```
flowchart LR
    U[User ticker] --> P[Planner]
```

```
    L[Research memory] --> P
    P --> T[Yahoo and SEC tools]
    P --> C[News prompt chain]
    Y[Yahoo Finance] --> T
    S[SEC EDGAR] --> T
    Y --> C
    N[NewsAPI] --> C
    G[Google News RSS fallback] --> C
    C --> R[Specialist router]
    R --> E[Earnings specialist]
    R --> M[Market specialist]
    R --> W[News specialist]
    E --> V[Evaluator / optimizer]
    M --> V
    W --> V
    T --> Z[Synthesizer]
    V --> Z
    Z --> Q[Report validation]
    Q --> F[Final report and visuals]
    F --> L
```

1.3 3. Environment setup

Create and activate a virtual environment, install `requirements.txt`, start Ollama, and copy `.env.example` to `.env`. At minimum:

- `OLLAMA_MODEL=llama3.2` and a running `ollama serve` process;
- `SEC_USER_AGENT` containing an application name and monitored contact email;
- optional `NEWS_API_KEY` (Yahoo remains primary and Google RSS is a fallback);
- optional `DEFAULT_TICKER`, `MAX_NEWS_ARTICLES`, `EVALUATION_SCORE_THRESHOLD`, and `MAX_OPTIMIZER_ITERATIONS`.

Credentials are read only from environment variables and are never displayed below.

[1]:
```python
import getpass
import os
import shutil
import subprocess
import sys
import time
import urllib.request
from pathlib import Path

try:
    import google.colab  # noqa: F401

    IN_COLAB = True
except ImportError:
    IN_COLAB = False
```

```
if IN_COLAB:
    repo_url = "https://github.com/bwirgau-usd/
 ↪AAI-520-final-project-multi-agent-financial-analysis.git"
    repo_dir = Path("/content/
 ↪AAI-520-final-project-multi-agent-financial-analysis")
    if not repo_dir.exists():
        subprocess.run(
            ["git", "clone", "--branch", "main", repo_url, str(repo_dir)],
            check=True,
        )
    os.chdir(repo_dir)
    subprocess.run(
        [sys.executable, "-m", "pip", "install", "-q", "-r", "requirements.
 ↪txt"],
        check=True,
    )

    if not os.getenv("NEWS_API_KEY"):
        news_key = getpass.getpass(
            "NEWS_API_KEY (optional; input hidden): "
        ).strip()
        if news_key:
            os.environ["NEWS_API_KEY"] = news_key
    if not os.getenv("SEC_USER_AGENT"):
        os.environ["SEC_USER_AGENT"] = input(
            "SEC user agent (name and contact email): "
        ).strip()

    if shutil.which("ollama") is None:
        subprocess.run(
            ["apt-get", "install", "-y", "-qq", "zstd"],
            check=True,
        )
        installer = urllib.request.urlopen(
            "https://ollama.com/install.sh",
            timeout=60,
        ).read()
        subprocess.run(["sh"], input=installer, check=True)
    subprocess.Popen(
        ["ollama", "serve"],
        stdout=subprocess.DEVNULL,
        stderr=subprocess.DEVNULL,
    )
    time.sleep(5)
    subprocess.run(
        ["ollama", "pull", os.getenv("OLLAMA_MODEL", "llama3.2")],
```

```
        check=True,
    )
    print("Colab environment ready.")
else:
    print("Local environment detected; start Ollama separately with ollama␣
 ↪serve.")
```

```
Local environment detected; start Ollama separately with ollama serve.
```

[2]:
```
import os
import sys
from pathlib import Path

import pandas as pd
from IPython.display import JSON, Markdown, display
from dotenv import load_dotenv


def find_project_root(start: Path) -> Path:
    """Find the repository root from JupyterLab or a nested notebook path."""
    for candidate in (start.resolve(), *start.resolve().parents):
        if (candidate / "src").is_dir() and (candidate / "requirements.txt").
 ↪is_file():
            return candidate
    raise FileNotFoundError("Could not locate the project root.")


PROJECT_ROOT = find_project_root(Path.cwd())
if str(PROJECT_ROOT) not in sys.path:
    sys.path.insert(0, str(PROJECT_ROOT))
load_dotenv(PROJECT_ROOT / ".env")

from src.data_sources.yahoo_finance import YahooFinanceClient
from src.reporting import (
    plot_10_year_stock_price,
    plot_news_analysis_summary,
    plot_news_workflow,
    plot_system_architecture,
    render_console_summary,
)
from src.tools.registry import TOOLS
from src.workflows import build_research_workflow

TICKER = os.getenv("DEFAULT_TICKER", "AAPL").strip().upper()
print(f"Project root: {PROJECT_ROOT}")
print(f"Ticker: {TICKER}")
print(f"NewsAPI configured: {bool(os.getenv('NEWS_API_KEY'))}")
```

```
print(f"SEC identity configured: {bool(os.getenv('SEC_USER_AGENT'))}")
```

```
Project root: /Users/brandonwirgau/Projects/AAI/520/Final Project
Ticker: AAPL
NewsAPI configured: False
SEC identity configured: False
```

1.4 4. Agent design, data sources, and tool use

Component	Implemented responsibility
Planner Agent	Uses the ticker, registered tools, and the last three relevant memories to produce objectives, questions, and an allow-listed tool plan.
Tool Executor	Runs only registered tools and isolates provider errors so one failure does not terminate the research run.
Market / Financial Tools	Compute price return, annualized volatility, maximum drawdown, company profile, valuation, statements, and cash flow.
SEC EDGAR Tool	Resolves CIK, reads official company facts, and lists recent 10-K, 10-Q, and 8-K filings with source metadata.
News Prompt Chain	Combines Yahoo and NewsAPI; invokes Google RSS when primary coverage is missing or sparse; deduplicates and structures articles.
Specialist Router	Sends classified items to earnings, market, or general-news specialists.
Evaluator / Optimizer	Scores observations, interpretation, uncertainty, and evidence breadth; revises until the threshold or iteration limit.
Synthesis Agent	Produces a grounded Markdown report spanning market, financial, filing, and news evidence.
Memory Curator / Store	Saves bounded, timestamped lessons using atomic file replacement; memory is planning context, never current evidence.

FRED and Alpha Vantage packages remain deliberate extension points. The report does not claim they are integrated data sources.

[3]:
```
tool_rows = [
    {
        "tool": name,
        "role": {
            "price_data": "One-year performance and risk",
            "company_info": "Profile, valuation, margins, and dividend",
            "financials": "Income-statement trends",
```

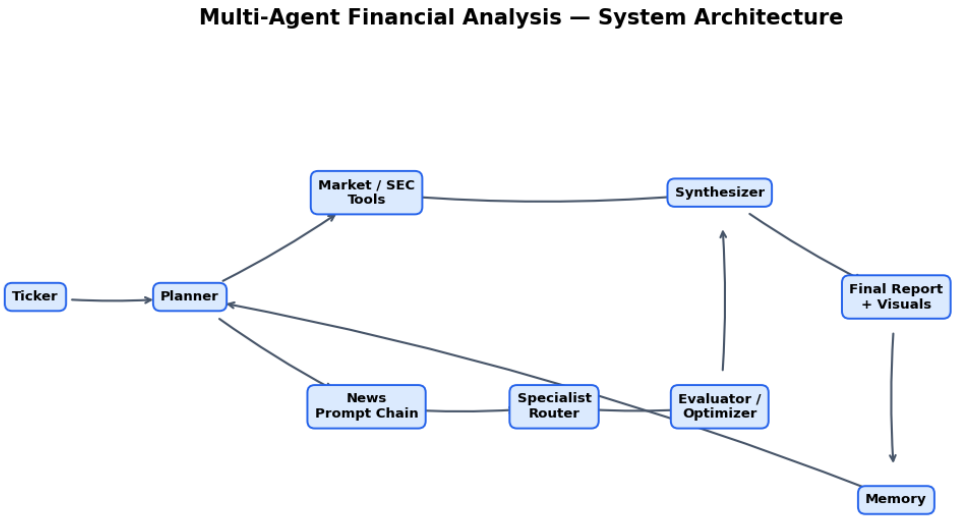
```
            "cash_flow": "Operating cash flow, free cash flow, and capex",
            "sec_edgar": "Official facts and recent filings",
        }.get(name, "Research evidence"),
    }
    for name in TOOLS
]
display(pd.DataFrame(tool_rows))
```

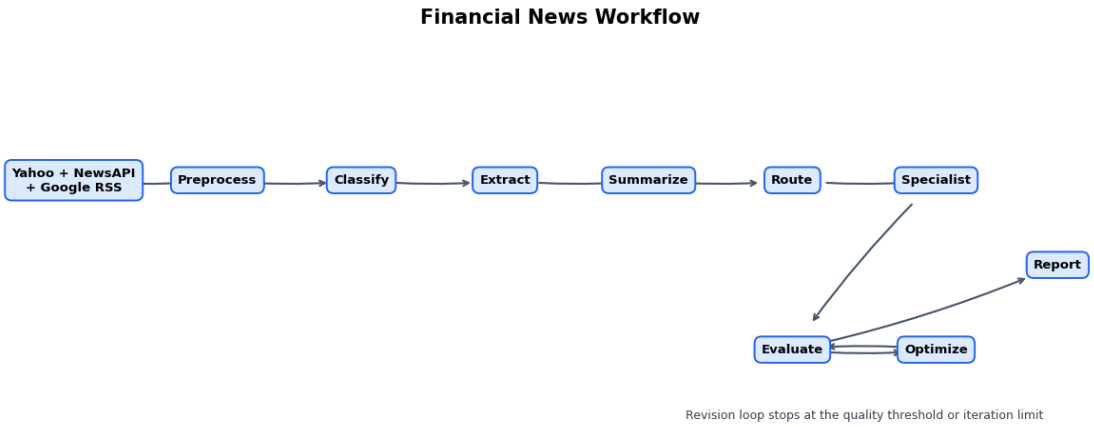
	tool	role
0	price_data	One-year performance and risk
1	company_info	Profile, valuation, margins, and dividend
2	financials	Income-statement trends
3	cash_flow	Operating cash flow, free cash flow, and capex
4	sec_edgar	Official facts and recent filings

1.5 5. Workflow diagrams

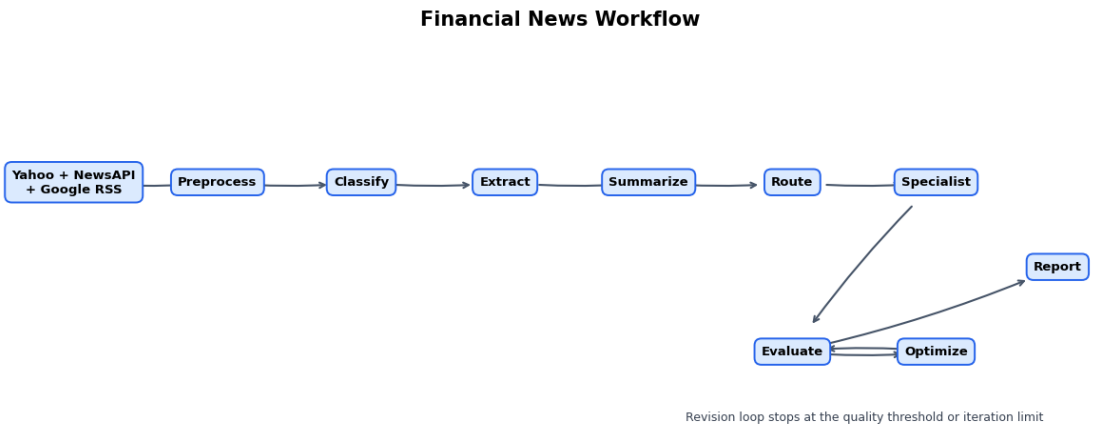
These diagrams are generated by the project's reporting module, so the submitted report reflects the implemented code rather than a disconnected illustration.

[4]:
```
plot_system_architecture()
plot_news_workflow()
```





[4]:



1.6 6. Prompt chaining, routing, and evaluator–optimizer logic

The updated news workflow uses provider isolation and explicit fallback logic:

```
flowchart LR
    Y[Yahoo Finance] --> D[Deduplicate and newest-sort]
    N[NewsAPI] --> D
    Y --> F{Coverage sparse?}
    N --> F
    F -->|Yes| G[Google News RSS]
    G --> D
    D --> P[Preprocess]
    P --> C[Classify]
    C --> X[Extract facts, figures, events]
    X --> S[Structured summary]
    S --> R{Route by category}
```

```
R --> E[Earnings]
R --> M[Market]
R --> W[General news]
E --> V[4 x 25-point evaluation]
M --> V
W --> V
V -->|Below threshold| O[Optimizer]
O --> V
V -->|Pass or max iterations| Z[Synthesis]
```

Provider failures are retained in `source_errors`; original and optimized analyses plus every evaluation score remain in workflow state for auditability. NewsAPI and RSS use retry-enabled HTTP sessions for transient 429/5xx and connection failures.

1.7 7. End-to-end investment research run

The next cell performs the live eight-stage workflow:

1. plan research;
2. collect Yahoo Finance and SEC evidence;
3. retrieve and process company news;
4. validate observations;
5. reflect on evidence quality;
6. synthesize the report;
7. validate report safety and consistency; and
8. save a reusable memory.

A running local Ollama model and network access are required for a complete live result. Missing optional providers are recorded rather than treated as fatal.

[5]:
```
workflow = build_research_workflow(project_root=PROJECT_ROOT)
result = workflow.run(TICKER, progress=print)
```

```
1/8 Planning the research run
2/8 Collecting market and financial evidence
Running tool: company_info
```

```
/Users/brandonwirgau/.local/share/virtualenvs/brandonwirgau-
gZAqjFTS/lib/python3.13/site-packages/requests/__init__.py:113:
RequestsDependencyWarning: urllib3 (2.7.0) or chardet (7.4.3)/charset_normalizer
(3.4.4) doesn't match a supported version!
  warnings.warn(
```

```
Running tool: financials
Running tool: price_data
Running tool: sec_edgar
3/8 Processing current company news
Processing news article 1/8
Processing news article 2/8
Processing news article 3/8
Processing news article 4/8
```

```
Processing news article 5/8
Processing news article 6/8
Processing news article 7/8
Processing news article 8/8
4/8 Validating tool observations
5/8 Reflecting on evidence quality
6/8 Synthesizing the research report
7/8 Validating the research report
8/8 Saving lessons for a future run
```

1.7.1 7.1 Structured planning and quality artifacts

[6]:
```
display(JSON({
    "symbol": result["symbol"],
    "plan": result.get("plan", {}),
    "validation": result.get("validation", []),
    "reflection": result.get("reflection", {}),
    "report_validation": result.get("report_validation", []),
    "memory_entry": result.get("memory_entry", {}),
}))
```

```
<IPython.core.display.JSON object>
```

1.7.2 7.2 Collected market, fundamental, and SEC evidence

The complete observation object is kept inspectable. Provider errors appear alongside successful results instead of being hidden.

[7]:
```
display(JSON(result.get("observations", {}), expanded=False))
```

```
<IPython.core.display.JSON object>
```

1.7.3 7.3 Detailed news analysis

This table surfaces source provenance, routing decisions, final evaluator scores, and optimizer activity. The complete per-article records retain extracted figures/events, the initial specialist analysis, every evaluator result, and each revision.

[8]:
```
news_analysis = result.get("news_analysis", {})
news_rows = []
for index, article in enumerate(news_analysis.get("articles", []), start=1):
    evaluation = article.get("evaluation", {})
    news_rows.append({
        "article": index,
        "source": article.get("source"),
        "published_at": article.get("published_at"),
        "headline": article.get("title"),
        "category": article.get("category"),
        "specialist": article.get("specialist"),
```

```
        "score": evaluation.get("score"),
        "passed": evaluation.get("passed"),
        "optimizer_revisions": len(article.get("optimization_history", [])),
        "url": article.get("url"),
    })

display(pd.DataFrame(news_rows))
display(JSON({
    "pipeline_stages": news_analysis.get("pipeline_stages", []),
    "retrieved_at": news_analysis.get("retrieved_at"),
    "retrieved_counts": news_analysis.get("retrieved_counts", {}),
    "source_errors": news_analysis.get("source_errors", {}),
    "source_counts": news_analysis.get("source_counts", {}),
    "routing_counts": news_analysis.get("routing_counts", {}),
    "average_score": news_analysis.get("average_score"),
    "passed_count": news_analysis.get("passed_count", 0),
}))
```

```
   article           source               published_at  \
0        1  Google News RSS  2026-10-15T22:30:43+00:00
1        2  Google News RSS  2026-10-15T20:11:26+00:00
2        3  Google News RSS  2026-10-15T16:15:59+00:00
3        4  Google News RSS  2026-10-15T13:38:15+00:00
4        5  Google News RSS  2026-10-15T10:19:03+00:00
5        6  Google News RSS  2026-10-15T08:38:41+00:00
6        7  Google News RSS  2026-10-15T07:20:18+00:00
7        8  Google News RSS  2026-10-13T12:00:02+00:00

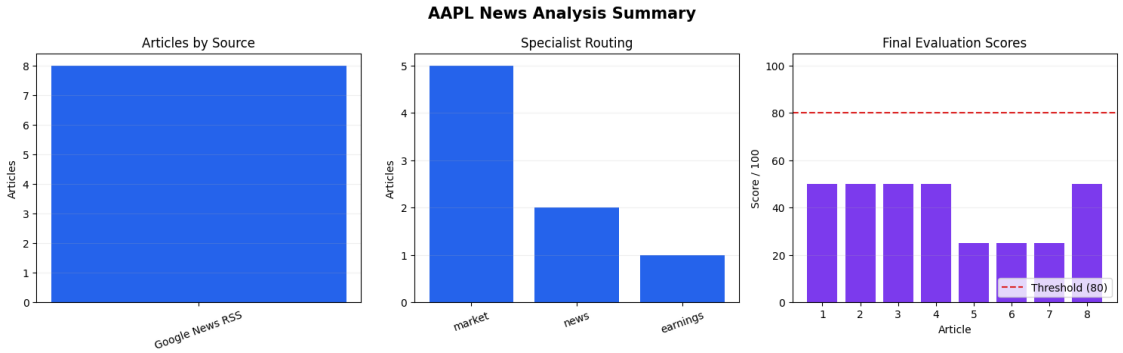
                                        headline  category specialist  \
0  Apple (AAPL) general counsel sells 2,399 share…    market     market
1  Stealth Bear Market Highlights Market Divergen…    market     market
2  Apple Has A Consumer Problem Nobody Talks Abou…      news       news
3  Apple (AAPL): Can Longer iPhone Ownership Stil…  earnings   earnings
4  Premarket Movers: AAPL Gains On China AI Appro…    market     market
5  Apple Inc. $AAPL Stock Position Boosted by Str…    market     market
6  Apple Inc. $AAPL Stock Acquired by Congress As…    market     market
7  Apple Inc. (AAPL) is Attracting Investor Atten…      news       news

   score  passed  optimizer_revisions  \
0     50   False                    3
1     50   False                    3
2     50   False                    3
3     50   False                    3
4     25   False                    3
5     25   False                    3
6     25   False                    3
7     50   False                    3
```

```
                                                  url
0  https://news.google.com/rss/articles/CBMipgFBV…
1  https://news.google.com/rss/articles/CBMiwAFBV…
2  https://news.google.com/rss/articles/CBMikgFBV…
3  https://news.google.com/rss/articles/CBMioAFBV…
4  https://news.google.com/rss/articles/CBMi5wFBV…
5  https://news.google.com/rss/articles/CBMizgFBV…
6  https://news.google.com/rss/articles/CBMivwFBV…
7  https://news.google.com/rss/articles/CBMioAFBV…

<IPython.core.display.JSON object>
```

[9]:
```
news_figure = plot_news_analysis_summary(news_analysis)
if news_figure is None:
    display(Markdown("*No news-summary chart is available because no articles␣
 ↪were retrieved.*"))
```



1.8 8. Final research report

The model-generated report uses only collected observations and explicitly distinguishes evidence, interpretation, risk, uncertainty, and missing data.

[10]:
```
display(Markdown(result["report"]))
```

2 Company Overview

- Company Name: Apple Inc.
- Symbol: AAPL
- Sector: Technology
- Industry: Consumer Electronics
- Country: United States
- Current Price: $342.0
- Market Capitalization: $4,991,209,307,376
- Enterprise Value: $5,013,154,299,904

3 Price Performance

- Start Price: $246.54
- Latest Price: $342.0
- 1-Year Return: 38.72%
- Annualized Volatility: 24.46%
- Maximum Drawdown: -13.8%
- Price Observation Count: 251

4 Valuation and Profitability

- Trailing P/E: 39.130436
- Forward P/E: 35.707386
- Price/ Sales: 10.691868
- Profit Margins: 0.27618998
- Operating Margins: 0.32623002
- Return on Equity: 1.4875101
- Beta: 1.069
- Dividend Yield: 0.32

5 Financial Performance

- Total Revenue (2025-09-30 00:00:00): $416,161,000,000.0
- Operating Income (2025-09-30 00:00:00): $133,050,000,000.0
- Net Income (2025-09-30 00:00:00): $112,010,000,000.0
- EBITDA (2025-09-30 00:00:00): $144,748,000,000.0
- Diluted EPS (2025-09-30 00:00:00): $7.46
- EBITDA/Net Income Ratio: 1.280

6 Cash Flow

- Operating Cash Flow: Not supplied
- Free Cash Flow: Not supplied
- Capital Expenditure: Not supplied
- Dividend Yield: 0.32

7 SEC EDGAR Evidence

- SEC Entity Name: Not supplied
- CIK: Not supplied
- SIC Classification: Not supplied
- Recent 10-K, 10-Q, and 8-K links: Not supplied

8 Dividend

- Dividend Yield: 0.32

9 Recent News and Catalysts

- Apple Inc. (AAPL) general counsel sells 2,399 shares at $345 each (Source: Google News RSS, 2026-10-15T22:30:43+00:00)
- Stealth Bear Market Highlights Market Divergence as Apple Inc (A - GuruFocus (Source: Google News RSS, 2026-10-15T20:11:26+00:00)
- Apple Has A Consumer Problem Nobody Talks About (NASDAQ:AAPL) - Seeking Alpha (Source: Google News RSS, 2026-10-15T16:15:59+00:00)
- Premarket Movers: AAPL Gains On China AI Approval Report – Why WMT, NFLX Stocks Are Also Rising - TradingView (Source: Google News RSS, 2026-10-15T10:19:03+00:00)

10 Risks and Uncertainties

- High debt-to-equity ratio
- High price-to-sales ratio
- High interest expenses
- RuntimeError: SEC_USER_AGENT is required and should identify the application and a monitored contact email.

11 Data Quality

- No data quality issues were identified.

12 Further Research

- Impact of interest rates on Apple's cash flow and valuation
- Estimated intrinsic value using discounted cash flow analysis

12.0.1 8.1 Deterministic report view

This second rendering guarantees that the core company overview, price performance, valuation, financial performance, cash flow, dividend, SEC filing evidence, news-source coverage, validation findings, and missing information remain visible even if model prose varies between runs.

[11]:
```
print(render_console_summary(result))
```

```
======================================================================
FINAL RESEARCH REPORT
======================================================================

Apple Inc. (AAPL)
----------------------------------------------------------------------

1. COMPANY OVERVIEW
   Sector:              Technology
   Industry:            Consumer Electronics
   Country:             United States
   Current Price:       $342.00
```

```
   Market Cap:          $4.991T
   Enterprise Value:    $5.013T

2. PRICE PERFORMANCE
   Start Price:         $246.54
   Latest Price:        $342.00
   One-Year Return:     38.72%
   Annualized Volatility: 24.46%
   Maximum Drawdown:    -13.80%
   Observations:        251

3. VALUATION
   Trailing P/E:        39.13
   Forward P/E:         35.71
   Price/Sales (TTM):   10.69
   Profit Margin:       27.62%
   Operating Margin:    32.62%
   Return on Equity:    148.75%
   Beta:                1.069

4. FINANCIAL PERFORMANCE (2025)
   Revenue:             $416.161B
   Operating Income:    $133.050B
   Net Income:          $112.010B
   EBITDA:              $144.748B
   Diluted EPS:         $7.46
   EBITDA / Net Income: 1.292x

5. CASH FLOW (2025)
   Operating Cash Flow: N/A
   Free Cash Flow:      N/A
   Capital Expenditure: N/A

6. SEC EDGAR EVIDENCE
   Unavailable:          RuntimeError: SEC_USER_AGENT is required and should
identify the application and a monitored contact email.

7. DIVIDEND
   Dividend Yield:      32.00%

8. RECENT NEWS AND CATALYSTS
   Retrieved At:        2026-10-16T03:27:31.819050+00:00
   Pipeline Stages:     ingest → preprocess → classify → extract → summarize →
route → specialist → evaluate → optimize
   Articles Processed:  8
   Retrieved Counts:    Yahoo Finance=0, NewsAPI=0, Google News RSS=8
   Source Counts:       Google News RSS=8
   Routing Counts:      market=5, news=2, earnings=1
```

```
   Average Score:       40.6/100
   Passed Evaluations:  0
   Score Threshold:     80.0/100
   Optimizer Limit:     3
   Article 1:
      Headline: Apple (AAPL) general counsel sells 2,399 shares at $345 each -
Stock Titan
      Source: Google News RSS / Stock Titan
      Published: 2026-10-15T22:30:43+00:00
      URL: https://news.google.com/rss/articles/CBMipgFBVV95cUxQOW5pLVlhOGg5ZGRG
LS1DcHJQZUNmVXpyZW5FQnkwem1VdjF5QW5qR3BpbUJlSzViVGlicHhNV2plekNGQnNrcHlELUN2MWhS
cU92REhrdjVjQW9EZkhGRXRjR0N6LWFpNHJQV1lyNktWdVNaallKVl9qeFdIaOFKTlVMM3puVVZidExV
eG8xYy11aHVJTlRrMWpic1AyTmhJT3pDRWNR?oc=5
      Route: market → market
      Events: none detected
      Figures: $345
      Keywords: shares, stock
      Evaluation: 50.0/100; passed=False
      Feedback: Add a clear interpretation.
      Feedback: Identify uncertainty or missing information.
      Observation: Market keywords: shares, stock
      Observation: Financial figures: $345
      Improvement: Add a clear interpretation
   Article 2:
      Headline: Stealth Bear Market Highlights Market Divergence as Apple Inc (A
- GuruFocus
      Source: Google News RSS / GuruFocus
      Published: 2026-10-15T20:11:26+00:00
      URL: https://news.google.com/rss/articles/CBMiwAFBVV95cUxQVFVhZ19xVml4N3pn
QlVGNW53YjFKWmlLQlp4WWp5YXBpcVNBRXFVVGRHMEp4TWdhOGVHV1RqTlNiOVI2ZEh5eDNmRUNLV0VY
SFFmZFN5V3ctS0xkYVBOMXNxa3c5ZG1UdUpIUEVTRWR1bUJoNHh4bXhkUVdDVDdFVVlKcFRhVlBJQlJm
a19HblJTNlhHX0JwOTNSaTJTRHczaUd5UXZlYnRnR2FtT053dXZVczBVTDZfalBoXzl3OGY?oc=5
      Route: market → market
      Events: none detected
      Figures: none detected
      Keywords: market
      Evaluation: 50.0/100; passed=False
      Feedback: Add a clear interpretation.
      Feedback: Identify uncertainty or missing information.
      Observation: Market keywords: market
      Observation: Financial figures: none
      Improvement: Add a clear interpretation.
      Improvement: Identify uncertainty or missing information.
   Article 3:
      Headline: Apple Has A Consumer Problem Nobody Talks About (NASDAQ:AAPL) -
Seeking Alpha
      Source: Google News RSS / Seeking Alpha
      Published: 2026-10-15T16:15:59+00:00
```

URL: https://news.google.com/rss/articles/CBMikgFBVV95cUxNbDJnQ3dBYWI2Z2k4eXJVZEVEczZ1M3BQOWY4WEpkVVFZSU1QSTZiWTE5SXpJZFRxd3FGNExBNk5yZWZkcER6cGN0MFVMbUxOZTFfb3lkMkNHZzd5MFhvUlRTczVGX3c3UWhJeW9zdXZyTm54QVBOU0VWakNsTEp3OWp1LS02cmtVSkRCX19oYVVlQQ?oc=5

Route: news → news

Events: none detected

Figures: none detected

Keywords: none detected

Evaluation: 50.0/100; passed=False

Feedback: Add a clear interpretation.

Feedback: Identify uncertainty or missing information.

Observation: Events: none

Observation: Keywords: none

Improvement: Add a clear interpretation.

Improvement: Identify uncertainty or missing information.

Article 4:

Headline: Apple (AAPL): Can Longer iPhone Ownership Still Produce More Profit per Customer? - Yahoo Finance

Source: Google News RSS / Yahoo Finance

Published: 2026-10-15T13:38:15+00:00

URL: https://news.google.com/rss/articles/CBMioAFBVV95cUxOVHRtOXNvNXNlN3VZbVUydFBpOVRaMjJXR1RlNk14M254QmdSMGlLcnRjamlibWthSXFxWlFQczNXY2xsV1JEWWxscm1tdE8zZGJiWmxzazJVd1pndXlHQXNDanlab0hvQlRBSm1kMTlXbzBEbDlYUkJVOTVtQnFpbXN0elZHblV4bmhyLVFVTHZHWXZ1X0k2MHdNVzZwbzBi?oc=5

Route: earnings → earnings

Events: none detected

Figures: none detected

Keywords: profit

Evaluation: 50.0/100; passed=False

Feedback: Add a clear interpretation.

Feedback: Identify uncertainty or missing information.

Observation: Financial figures: none

Observation: Events: none

Improvement: Add a clear interpretation.

Improvement: Identify uncertainty or missing information.

Article 5:

Headline: Premarket Movers: AAPL Gains On China AI Approval Report - Why WMT, NFLX Stocks Are Also Rising - TradingView

Source: Google News RSS / TradingView

Published: 2026-10-15T10:19:03+00:00

URL: https://news.google.com/rss/articles/CBMi5wFBVV95cUxPRlFOMDNmcUdwMTItLTZ5UWhCaEhsTzBrcVpRSW5ZUVZndXlfY1lMRHZicmVoNS1rYnJYajdKRVFEYUhudlE4SFlmY0VscndnRlFrWGpHZmsxTEhoN21WUkptWExNSWljT1M3QzFFVEpiSDY5c0V5TmRzamJtTnR2cWEyV1RLUVZoOGk2TXdCWWJSQmpMRHFlWDJJeHF2azA4Um0yRkZyQkZDVGFvYWZEandlQWVMMExXdUxxYWJoWDduS0xyVXd3eVRzd0k1R1M5SDB5cUlmdTRFc0ZkX1p0dDdCM0Nhc3c?oc=5

Route: market → market

Events: none detected

Figures: none detected

Keywords: market, stock, AI
Evaluation: 25.0/100; passed=False
Feedback: Add a clear interpretation.
Feedback: Identify uncertainty or missing information.
Feedback: Include at least two evidence-based observations.
Observation: Market keywords: market, stock, AI
Improvement: Add a clear interpretation.
Improvement: Identify uncertainty or missing information.

Article 6:

Headline: Apple Inc. $AAPL Stock Position Boosted by Strong Retirement Solutions LLC - MarketBeat
Source: Google News RSS / MarketBeat
Published: 2026-10-15T08:38:41+00:00
URL: https://news.google.com/rss/articles/CBMizgFBVV95cUxOV3lqR2thLUptcTk1RE1iSjBnUjZXYVVyUFZjWkFmamQ0SXRoN1pFcnluRVBsR3BRaHZCNGp6dUdiUl94bDRvY3RoYzdFd3lLOFM5UGN4YUwtYUN2NEx0ejc1RmJWMWV2b2ZRdGZLUjBrUEJXYlhOVERIbjZyMGlxUHoxU1c0TUlRV01HbTZoNWVPUnJWWjkxZjU1Rk5hVFA3RzBsbWdxVXhFemVNblRrU2NFM1I5bEJMWFMxUWJVS25kUmJFYTBPd0hhcWZJUQ?oc=5
Route: market → market
Events: none detected
Figures: none detected
Keywords: market, stock
Evaluation: 25.0/100; passed=False
Feedback: Add a clear interpretation.
Feedback: Identify uncertainty or missing information.
Feedback: Include at least two evidence-based observations.
Observation: Market keywords: market, stock
Improvement: Add a clear interpretation.
Improvement: Include at least two evidence-based observations

Article 7:

Headline: Apple Inc. $AAPL Stock Acquired by Congress Asset Management Co. - MarketBeat
Source: Google News RSS / MarketBeat
Published: 2026-10-15T07:20:18+00:00
URL: https://news.google.com/rss/articles/CBMivwFBVV95cUxQcTRzM3Y2dVJSOEJ3WGo3dTByX3kzQ2IwSDc2MUhGX2dTQVVybmZzX1VRVWRVbHFPMEo0OWhDV1JkMFFRdXhBMlNGMnp2ckY5TlEta09jXzQ0UzhrOHdFalVTODhUUlVGaEpiZ3o4M2tmZjRBUWdmN1p2U29ISElIc1RnakhYVFJ2S1dzZHp3Y2RiNlhzRHhoMUFQVDIwX1hOWlNwZzFlTjU3aGxiVWhxUmVuNUs3am94d0VyM3Z0OA?oc=5
Route: market → market
Events: none detected
Figures: none detected
Keywords: market, stock
Evaluation: 25.0/100; passed=False
Feedback: Add a clear interpretation.
Feedback: Identify uncertainty or missing information.
Feedback: Include at least two evidence-based observations.
Observation: Market keywords: market, stock
Improvement: Add a clear interpretation.

```
      Improvement: Identify uncertainty or missing information.
   Article 8:
      Headline: Apple Inc. (AAPL) is Attracting Investor Attention: Here is What
You Should Know - Yahoo Finance
      Source: Google News RSS / Yahoo Finance
      Published: 2026-10-13T12:00:02+00:00
      URL: https://news.google.com/rss/articles/CBMioAFBVV95cUxQZXRvVGFmMjJWQOQz
MFBvd1JjOTRJalphVXJnWmM2Q25WMmFxQzN6UFpFbU52bXhrQzF2bXR5anplajk4QUMxNHBIX25zVy1F
c3V0aDRWMU55WlNUdUhfX3Rfd2M1eWRIYmxzOXRaRkttcE95UTlvQ0hMcEE3UFFaSFlWV3ZxUVNfU1NZ
RF83dTlnMkh4dnE4ZUp5dVMtMnlw?oc=5
      Route: news → news
      Events: none detected
      Figures: none detected
      Keywords: none detected
      Evaluation: 50.0/100; passed=False
      Feedback: Add a clear interpretation.
      Feedback: Identify uncertainty or missing information.
      Observation: Events: none
      Observation: Keywords: none
      Improvement: Add a clear interpretation.
      Improvement: Identify uncertainty or missing information.

9. RISKS AND UNCERTAINTIES
   - High interest expenses
   - RuntimeError: SEC_USER_AGENT is required and should identify the
application and a monitored contact email.
   - High debt-to-equity ratio
   - High price-to-sales ratio

10. DATA QUALITY
   - Evidence strength: High debt-to-equity ratio
   - Evidence strength: High price-to-sales ratio
   - sec_edgar: RuntimeError: SEC_USER_AGENT is required and should identify the
application and a monitored contact email.

11. MISSING INFORMATION
   - Interest rates

12. FURTHER RESEARCH
   - Impact of interest rates on Apple's cash flow and valuation
   - What is Apple's current debt structure and interest rates?
   - How do interest expenses impact Apple's cash flow and valuation?
   - What are the financial performance metrics for Apple and its industry
peers?
   - What are the estimated intrinsic values of Apple using discounted cash flow
analysis?
   - How does Apple's financial performance compare to its industry peers?
```

```
================================================================================
END OF FINAL RESEARCH REPORT
================================================================================
```

12.1 9. Ten-year market context

The original notebook's 10-year graph is preserved as a reusable reporting function. It uses adjusted closing prices and reports the start price, end price, and total return.

[12]:
```python
try:
    price_chart = plot_10_year_stock_price(YahooFinanceClient(), TICKER)
    if price_chart is None:
        display(Markdown(f"*No 10-year price history was available for {TICKER}.*"))
    else:
        print(price_chart.summary)
except Exception as exc:
    display(Markdown(f"*The 10-year chart could not be loaded: {type(exc).__name__}: {exc}*"))
```




```
10-year price summary: $26.88 → $342.00 (+1,172.2%)
```

12.2 10. Evaluation evidence and iteration

Each news analysis receives 25 points for each of the following:

- at least one concrete observation;
- a clear interpretation;
- an explicit uncertainty or missing-information statement; and
- at least two evidence-based observations.

Below-threshold work is revised up to `MAX_OPTIMIZER_ITERATIONS`. Deterministic observation validation separately checks missing tools/metrics, nonpositive prices, implausible percentage scales, market-cap/revenue scale mismatches, and cash-flow consistency. Final-report validation flags false missing-data claims and prescriptive or misleading investment language.

[13]:
```
iteration_rows = []
for index, article in enumerate(news_analysis.get("articles", []), start=1):
    for revision, evaluation in enumerate(article.get("evaluation_history",␣
 ↪[])):
        iteration_rows.append({
            "article": index,
            "revision": revision,
            "score": evaluation.get("score"),
            "passed": evaluation.get("passed"),
            "feedback": "; ".join(evaluation.get("feedback", [])),
        })
display(pd.DataFrame(iteration_rows))
```

	article	revision	score	passed	\
0	1	0	50	False	
1	1	1	50	False	
2	1	2	50	False	
3	1	3	50	False	
4	2	0	50	False	
5	2	1	50	False	
6	2	2	50	False	
7	2	3	50	False	
8	3	0	50	False	
9	3	1	50	False	
10	3	2	50	False	
11	3	3	50	False	
12	4	0	50	False	
13	4	1	50	False	
14	4	2	50	False	
15	4	3	50	False	
16	5	0	50	False	
17	5	1	50	False	
18	5	2	25	False	
19	5	3	25	False	
20	6	0	50	False	
21	6	1	25	False	
22	6	2	25	False	
23	6	3	25	False	
24	7	0	25	False	
25	7	1	25	False	
26	7	2	25	False	
27	7	3	25	False	
28	8	0	50	False	

```
29        8         1     50   False
30        8         2     50   False
31        8         3     50   False
```

```
                                                 feedback
0   Add a clear interpretation.; Identify uncertai…
1   Add a clear interpretation.; Identify uncertai…
2   Add a clear interpretation.; Identify uncertai…
3   Add a clear interpretation.; Identify uncertai…
4   Add a clear interpretation.; Identify uncertai…
5   Add a clear interpretation.; Identify uncertai…
6   Add a clear interpretation.; Identify uncertai…
7   Add a clear interpretation.; Identify uncertai…
8   Add a clear interpretation.; Identify uncertai…
9   Add a clear interpretation.; Identify uncertai…
10  Add a clear interpretation.; Identify uncertai…
11  Add a clear interpretation.; Identify uncertai…
12  Add a clear interpretation.; Identify uncertai…
13  Add a clear interpretation.; Identify uncertai…
14  Add a clear interpretation.; Identify uncertai…
15  Add a clear interpretation.; Identify uncertai…
16  Add a clear interpretation.; Identify uncertai…
17  Add a clear interpretation.; Identify uncertai…
18  Add a clear interpretation.; Identify uncertai…
19  Add a clear interpretation.; Identify uncertai…
20  Add a clear interpretation.; Identify uncertai…
21  Add a clear interpretation.; Identify uncertai…
22  Add a clear interpretation.; Identify uncertai…
23  Add a clear interpretation.; Identify uncertai…
24  Add a clear interpretation.; Identify uncertai…
25  Add a clear interpretation.; Identify uncertai…
26  Add a clear interpretation.; Identify uncertai…
27  Add a clear interpretation.; Identify uncertai…
28  Add a clear interpretation.; Identify uncertai…
29  Add a clear interpretation.; Identify uncertai…
30  Add a clear interpretation.; Identify uncertai…
31  Add a clear interpretation.; Identify uncertai…
```

12.3 11. Memory and learning across runs

The memory subsystem stores concise lessons and weaknesses in `data/memory/research_memory.json`. Writes are atomic and bounded, lookups are case-insensitive, and only the last three ticker-relevant memories enter a future planning prompt. The planner is explicitly instructed not to treat old notes as current market evidence.

[14]:
```
display(JSON(result.get("memory_entry", {})))
```

```
<IPython.core.display.JSON object>
```

12.4 12. Limitations, responsible use, and conclusion

- Yahoo Finance, NewsAPI, Google News RSS, and SEC EDGAR can be delayed, unavailable, rate-limited, incomplete, or inconsistent.
- News summaries may contain only provider snippets; source links should be reviewed before decisions.
- Keyword extraction and routing are intentionally transparent but cannot capture every nuance.
- Local-model output is nondeterministic and may still omit context; deterministic validation reduces but cannot eliminate that risk.
- Google News RSS is a coverage fallback, not a replacement for primary or official evidence.
- FRED and Alpha Vantage remain scaffolds and are not represented as report evidence.
- Historical performance, evaluator scores, and model interpretations do not predict future returns.

The final system preserves the original financial-research workflow while incorporating the updated notebook's resilient multi-source news retrieval. Its modular adapters, explicit state, provenance, bounded revision loop, deterministic report, persistent memory, offline tests, and generated diagrams make the work reproducible and auditable.

12.5 Appendix — complete workflow state

Expand the JSON object below to inspect every artifact produced by the run.

[15]:
```
display(JSON(result, expanded=False))
```

```
<IPython.core.display.JSON object>
```